

RESEARCH EXPERIENCE

- Computational Imaging Lab | VISITING STUDENT

MAR 2025 – NOW

UC Berkeley, EECS

Working with Prof. Laura Waller, designing a computational spectrometer with information-driven deep learning frameworks and building a setup for optical diffraction tomography.

Ongoing Research: Info-driven Optimization, Multi-wave ODT
- Optical Imaging System Lab | PH.D CANDIDATE

MAR 2019 – NOW

Yonsei University, Electronic and Electric Engineering

Conducted research under the supervision of Prof. Seung Ah Lee in the Optical Imaging System Lab(now Imaging Intelligence Lab), focusing on lensless imaging techniques and computational imaging methods.

Ongoing Research: 3D Depth imaging, Task-Optimized PSF Design

RESEARCH PROJECTS

- Lensless 3D Depth Imaging | 2023 - Now

- Keywords: Depth Imaging, PSF Engineering, Wave optics, SLM

Design a lensless camera for 3D imaging, enriching depth resolution with optimized phase masks forming multiple depth-dependent point spread functions using a spatial light modulator.
- Lensless Snapshot Hyperspectral Imaging | 2021 - 2023

- Keywords: Hyperspectral Imaging, Linear variable filter, Calibration

Build a snapshot lensless imaging system that utilizes a linear variable filter and a phase mask to encode spectral information, enabling recovery of hyperspectral image stacks from a single measurement.
- Lensless High Speed Eye Tracking | 2019 - 2021

- Keywords: Eye tracking, Wearable device, Rolling-shutter

Design and construction of a goggle-type eye tracker using a low-cost and high-speed lensless camera with rolling shutter image sensor for monitoring eye movements in neurodegenerative diseases.

PUBLICATIONS

- JOURNAL

Kim, T., Lee, K. C., Baek, N., Chae, H., & Lee, S. A. (2023). Aperture-encoded snapshot hyperspectral imaging with a lensless camera. APL Photonics, 8(6).

Baek, N., Lee, Y., Kim, T., Jung, J., & Lee, S. A. (2022). Lensless polarization camera for single-shot full-Stokes imaging. APL Photonics, 7(11).

Lee, Y., Chae, H., Lee, K. C., Baek, N., Kim, T., Jung, J., & Lee, S. A. (2022). Fabrication of integrated lensless cameras via UV-imprint lithography. IEEE Photonics Journal, 14(2), 1-8.
- PROCEEDING

Kim, T., Lee, Y., Jung, J., Lee, K. C., & Lee, S. A. (2024, August). 3D Lensless Camera for extended depth range with multiple point spread functions. In Conference on Lasers and Electro-Optics/Pacific Rim (p. We3G_3). Optica Publishing Group.

Kim, T., Lee, K. C., Lee, K., Baek, N., Jung, J., Kim, E., ... & Lee, S. A. (2024, February). High-speed lensless eye tracker for microsaccade measurement. In SPIE Advanced Biophotonics Conference (SPIE ABC 2023) (Vol. 13076, pp. 38-44). SPIE.

- RECENT CONFERENCE ACTIVITY

Miniaturized Computational Spectrometer with Information-driven Design (Poster) (Photonics West 2026)

Privacy-Preserving Imaging With Lensless Cameras Using Shift-Variant Point Spread Functions (Oral) (COSI 2025)

Task-Optimized Lensless Imaging With Trainable Phase Masks for Compact Spatial Encoding (Oral) (COSI 2025)

Quantitative Fluorescence Imaging for Detection of Dental Plaque with Lensless Camera (Poster) (ABC 2024)

3D Lensless Camera for Extended Depth Range with Multiple Point Spread Functions (Oral) (CLEO-PR 2024)

EDUCATION

- YONSEI UNIVERSITY

MS & PH.D OF ENGINEERING IN SCHOOL OF ELECTRONIC AND ELECTRICAL ENGINEERING

Expected Sep 2026 | Republic of Korea, Seoul
- YONSEI UNIVERSITY

BACHELOR OF ENGINEERING IN SCHOOL OF ELECTRONIC AND ELECTRICAL ENGINEERING

Feb 2021 | Republic of Korea, Seoul

HONORS

- AWARDS

2024 ABC Best Poster Award, Optical Society of Korea (2024)

2023 SPIE-ABC Young Investigator Award, Optical Society of Korea (2023)

Outstanding Presentation Award, Capstone Design Project, (2019)

- PRESS

AIP Scilight, "Improving hyperspectral imaging using a lensless camera", 2023

- GRANTS

Selected Participant, BK21(Brain Korea 21) International Joint Training Program, National Research Foundation of Korea (NRF), 2025

Undergraduate Research Fellow, Undergraduate Research Program, Korea Science and Creativity Foundation (KOSAC), 2019

- SOCIETY

Regular Member, Yonsei EEHS(Honor Society)

PATENTS

- Apparatus and method for measuring eye movement - Lee, S. A., Kim, T. Y., Lee, K. W., Baek, N. K., & Jung, J. W. (2024). U.S. Patent Application No. 18/052,192.

Methods for manufacturing phase masks and lens-less camera module - Lee, S. A., Chae, H. S., Lee, Y. J., Lee, K. C., Baek, N. K., Kim, T. Y., & Jung, J. W. (2023). U.S. Patent Application No. 18/168,887.

SKILLS

- TECHNICAL

Python(Pytorch, OpenCV) • MATLAB • Fusion 360
- LANGUAGES

Korean (Native) • English (Intermediate, OPIc IM)

ADDITIONAL INFO

- For Other Projects / Publications / Patents / Conference Activities

